

GABRIEL WOMELSDORF

Newport News, Virginia · (410) 591-0511 · gabriel.womelsdorf.work@gmail.com
linkedin.com/in/gabriel-womelsdorf-038987224 · github.com/silverwarriorin · silverwarriorin.github.io

SUMMARY

Computer Engineering student (B.S. 2027) developing embedded firmware and custom hardware with C/C++, Zephyr RTOS/NCS, FreeRTOS, Nordic nRF52/nRF54, STM32, and Bluetooth Low Energy. Builds systems from architecture and PCB design through firmware, sensor integration, power sequencing, and board bring-up; brings 4+ years of production software experience progressing from intern to part-time lead engineer.

EDUCATION

Christopher Newport University

B.S., Computer Engineering, *expected 2027*; Minor, Mathematics

Newport News, VA
CNU Summer Scholars, 2026

TECHNICAL SKILLS

Firmware: C, C++, Zephyr RTOS/NCS, FreeRTOS, STM32 HAL, ESP-IDF, Python

Wireless & MCUs: Bluetooth Low Energy (BLE), Nordic nRF52/nRF54, STM32G0

Interfaces & Debug: I2C, SPI, UART, SWD, ADC/DMA, sensor integration

Hardware & Tools: PCB design (KiCad, Altium), board bring-up, USB-C Power Delivery, power sequencing, PMBus, Git, CMake

EMBEDDED SYSTEMS PROJECTS

- **Distributed BLE Sensing System & Production Test Platform (Capstone, in progress):** Architected a privacy-preserving multi-node sensing system and designed custom battery-powered node hardware around Nordic nRF54L15 BLE modules, nPM1300 power management, ADXL367 accelerometers, and SPI flash. Developing a four-layer STM32G0 test fixture to program and qualify up to 16 nodes, with 15-V USB-C PD, protected 5-V/3.3-V rails, I2C/SPI/UART/SWD, 84 test points, and FreeRTOS firmware for PD negotiation, ADC/DMA rail monitoring, sequenced power-up, fault shutdown, and debug logging.
- **Air Sentry Neo:** Designed and built a battery-powered indoor air-quality research platform end-to-end, including electronics and mechanical design, PCB layout and assembly, embedded firmware, and integration of five I2C sensors. Achieved approximately 3.5 weeks of operation from a 350 mAh LiPo; owns hardware qualification, experimental design, and manuscript development for active unpublished additive-manufacturing research.
- **Tardis Embedded-AI Development Kit:** Proposed and secured Core4ce-backed CNU Summer Scholars funding for a modular offline edge-AI platform. Owned architecture across a reusable compute module, carrier boards, embedded firmware, and developer tooling; independently assembled and brought up the prototype hardware.

ENGINEERING EXPERIENCE

Core4ce

Software R&D Intern → Software Engineer → Lead Software Engineer (part-time)

Reston, VA
Jun 2022–Present

- Led end-to-end development of Cyberscape, owning architecture decisions, sprint planning, client deployments, and mentorship of junior engineers.
- Designed and implemented a production Apache NiFi data pipeline on AWS, including custom Java processors, IAM-compliant integrations, and reconfigurable routing.
- Led a two-month testing overhaul for a previously untested subsystem, substantially increasing coverage and bringing the component to production-ready quality.

Christopher Newport University

Student Researcher (part-time)

Newport News, VA
Nov 2023–May 2025

- Explored CNC routing methods for fabricating table-sized PCBs for the Super BigByte project.
- Assembled, installed, and documented power-delivery systems for the coordinate detector's photomultipliers.

LEADERSHIP

The Captains Workshop, CNU — Founder and President

Feb 2023–Present

Founded CNU's 3D-printing club and organize hands-on workshops and shared technical resources.